PACE 2027

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https://pacechallenge.org/

Scientific Topic: Longest Path

Definition

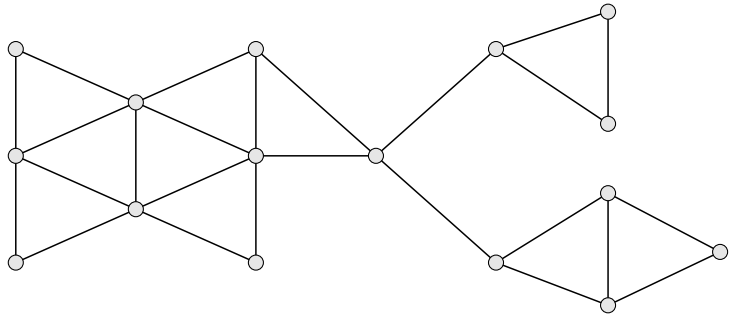
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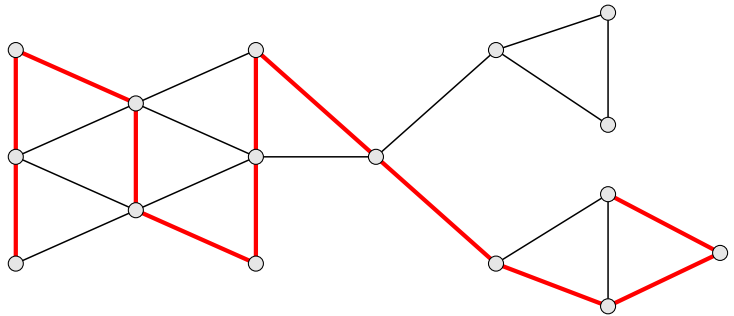
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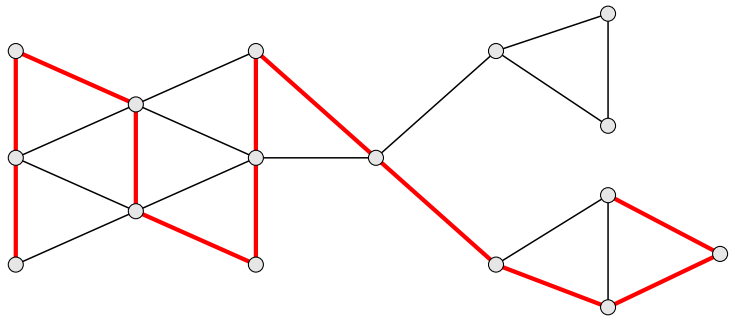
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Results for parameter k

- NP-hard — Karp '72
- randomized $O^*((2e)^k)$ — Alon, Yuster, Zwick '95
  randomized $O^*(1.66^k)$ — Björklund, Husfeldt, Kaski, Koivisto '17
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- no polynomial kernel unless NP $\subseteq$ coNP/poly — Bodlaender, Downey, Fellows, Hermelin '09

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Results for treewidth tw

- randomized $4^{\mathrm{tw}} n^{O(1)}$ — Cygan et al. '11
- deterministic $2^{O(\mathrm{tw})}$ — Bodlaender et al. '15
- Hamiltonicity: $(2+\sqrt{2})^{\mathrm{pw}}$ — Cygan, Kratsch, Nederlof '13

Organization and Data

Exact Unweighted Track

- **undirected, unweighted** graph G
- output: largest k such that G has a k-path
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Solver Review

- in the exact tracks, solvers with **one-sided error probability** are allowed
- pure heuristics will be disqualified
- participants may **inspect** other solvers

Timeline

mostly follows previous PACE-instances

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– good luck and an enjoyable competition –